Chocolate Ratings

Final Group Project, Group Number: 11

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Table 1: Group Members

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Course Name: Statistical Aspects of Modelling

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1. Introduction:

The dataset used in this project is titled "Chocolate Ratings", published as part of the TidyTuesday Project on January 18, 2022.The Chocolate Ratings dataset was originally compiled by Flavors of Cacao, a website dedicated to reviewing and rating artisanal chocolate bars from around the world.

Each observation in the dataset represents a single chocolate bar review. The data include information about the manufacturer, the country and region of bean origin, the cocoa percentage, and the expert rating on a 1–5 scale. The dataset contains approximately 2,500 observations and around 9 variables, offering a mix of quantitative and categorical data suitable for exploratory and inferential analysis. This dataset is particularly appealing for statistical modeling because it allows for the investigation of how chocolate characteristics affect expert evaluations.

- The dataset was obtained from the TidyTuesday GitHub repository maintained by the R for Data Science online learning community (2022).

2. Variable Descriptions:

Table 2: Variable Descriptions for Chocolate Ratings Dataset

Variable	Description	Type	Example	Use_in_Analysis
company_manufacturer	Name of the company that made the chocolate bar	Categorical	A. Morin	Context variable
review_date	Year the chocolate bar was reviewed	Numeric (integer)	2019	Time variable
country_of_bean_origin	Country where the cocoa beans were grown	Categorical	Peru	Alternative predictor
broad_origin	Broader region of bean origin	Categorical	South America	Predictor variable
cocoa_percent	Cocoa content of the chocolate bar, expressed as a percentage	Numeric (continuous)	70	Main predictor (X)
rating	Expert rating score (1.0–5.0 scale)	Numeric (continuous)	3.75	Response variable (Y)

Variable	Description	Type	Example	Use_in_Analysis
bean_type	Variety or hybrid of cocoa bean used	Categorical	Trinitario	Context variable
company_location	Country where the manufacturer is based	Categorical	France	Optional predictor

Key Variables of Interest:

For this project, the **response variable (Y)** and potential **predictor variables (X)** will be included:

Response Variable (Y): The Expert Rating(rating), a score between 1 and 5 representing how experts evaluate the quality of the chocolate.

Predictor Variable (X): The Cocoa Percentage(cocoa_percent) and Country of origin (country_of_bean_origin)

These predictors are well suited to **regression analysis** and **hypothesis test**, allowing exploration of whether cocoa concentration and bean origin explain variation in chocolate ratings.

3. Research Context:

Chocolate quality is influenced by many factors, including bean variety, origin, and processing methods. From a data-driven perspective, this dataset enables an investigation of whether measurable characteristics such as cocoa content and geographic origin systematically affect expert ratings.

Possible research questions include:

1. **Does the cocoa percentage predict the chocolate's expert rating?**
   – A simple linear regression can test whether higher cocoa content corresponds to higher quality ratings.
2. **Is the mean rating significantly different between chocolates with $\geq 70\%$ cocoa and those with <70% cocoa?**
   – A hypothesis test can compare in mean ratings between high- and low-cocoa chocolates.
3. **Do chocolates from South America have higher ratings than chocolates from Africa?** -A hypothesis test can compare whether the average rating differs between South America and Africa.

Suitability for analysis: The dataset contains a clear numeric response variable, rating, measured on a continuous 1–5 scale, and several quantitative predictors, such as cocoa_percent. Additionally, it includes categorical variables (e.g., country_of_bean_origin, company_location) that can be used to define groups or compare means. This combination allows for both simple and multiple linear regression analyses, as well as hypothesis testing across groups.

Sample size: With approximately 2,500 observations, the dataset is large enough to ensure stable parameter estimates and to detect statistically significant relationships where they exist. The sample size provides sufficient power for regression modeling and group comparisons.

Independence of observations: Each chocolate bar in the dataset is reviewed as an independent product. While multiple bars may come from the same company, it is reasonable to assume that ratings are made independently based on the characteristics of each bar, satisfying the independence assumption required for regression and hypothesis testing.

4. Data Quality Considerations:

The Chocolate Ratings dataset is relatively clean and well documented, but several data quality issues warrant consideration:

i. **Missing Values:** Some categorical variables such as `bean_type` or `broad_origin` contain missing or "unknown" entries. These can be handled by removing incomplete rows or grouping missing categories as "Other."

ii. **Data Cleaning for Cocoa Percentage:**
The `cocoa_percent` column is stored as a character string with a percentage sign (e.g.,`"70%"`). This must be converted to numeric format for statistical analysis.

iii. **Potential Outliers:** Ratings are on a 1–5 scale but typically cluster between 2.5 and 4.5. Visual inspection use to detect any extreme or unusual values.

iv. **Measurement Consistency:** Ratings were given by the same review organization, reducing bias, but subjectivity remains a limitation because taste perception varies.

v. **Data Collection Concerns:** The dataset also focuses mainly on premium chocolate brands, which may not represent the full global market. In addition, some variables such as bean_type and country_of_bean_origin contain missing or inconsistent data, which may limit sample size or affect accuracy. Ratings were collected over multiple years, so temporal differences in scoring standards could exist.

5. Methodology:

We have analyzed the Chocolate Ratings dataset using structured steps of data cleaning, exploratory data analysis, regression modeling, and hypothesis testing. Data prepossessing involved converting cocoa percentages to numeric form and addressing missing values. EDA included summary statistics and visualizations of rating distributions and cocoa–rating relationships. A simple linear regression assessed whether cocoa percentage predicts ratings, while Welch's two-sample t-test compared mean ratings between chocolates with $\geq 70\%$ and $<70\%$ cocoa.

6. Exploratory Data Analysis(EDA):

6.1 Basic summary about the dataset:

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.000   3.000   3.250   3.196   3.500   4.000
```

Interpretation: - The chocolate ratings are centered around 3.2, indicating most chocolates are rated fairly well.

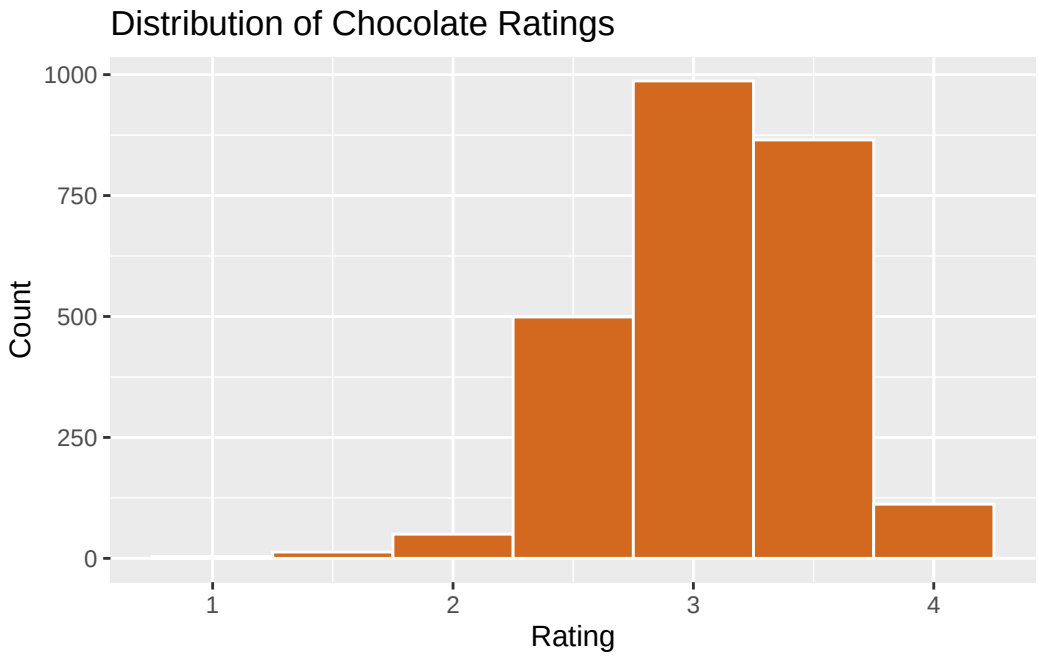
- Since mean (3.196) median (3.25), the distribution is roughly symmetric — not strongly skewed.
- Range $= 4 - 1 = 3$, so chocolate ratings vary moderately.

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  42.00   70.00   70.00   71.64   74.00  100.00
```

Interpretation: - Most chocolates have cocoa content around 70–74%, meaning they are mostly dark chocolates.

- The mean $>$ median, so the distribution might be slightly right-skewed, meaning a few very high-cocoa bars (e.g., 90–100%) pull up the average a bit.

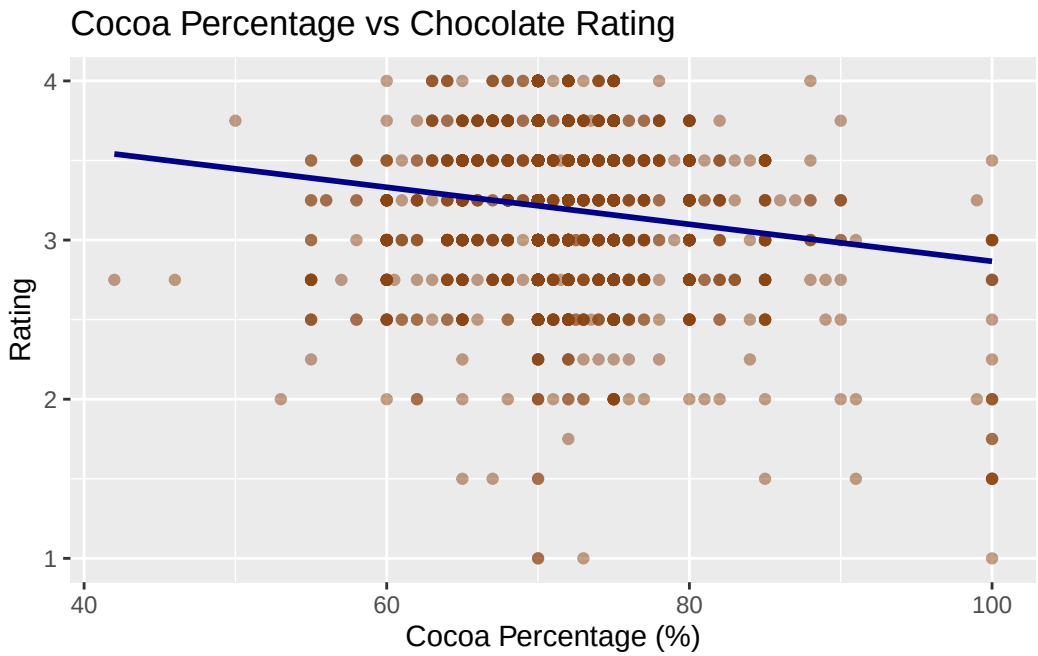
6.2 Distribution of Ratings:



Interpretation:

The data suggests that most chocolates in the dataset are rated around the middle (3 out of 5), with fewer chocolates being rated at the extremes (1, 2, 4, or 5).The majority of chocolates receive above-average ratings, implying that the dataset likely contains many well-reviewed products. In summary, the ratings tend to be centered around a moderate score, with fewer extreme ratings at either end. Since the distribution is slightly left-skewed (more high scores), it may indicate a positive bias in ratings (reviewers tend to rate chocolates favorably).

6.3 Relationship Between Cocoa and Rating:



Interpretation:

The scatter plot is a weak negative relationship between cocoa percentage and chocolate ratings. As cocoa content increases, ratings tend to decrease slightly, suggesting that chocolates with very high cocoa content are generally rated lower. However, the wide dispersion of points indicates that cocoa percentage alone does not strongly predict ratings-other factors, such as texture, flavor notes, or brand reputation, may also influence reviewer preferences.

7. Analysis Section:

7.1 Correlation:

Question-1: Does cocoa percentage predict the expert rating of chocolate bars?

simple linear regression:

A simple linear regression will be fitted to model the relationship between cocoa content and chocolate rating.

```
Call:
lm(formula = rating ~ cocoa_percent, data = chocolate)

Residuals:
     Min       1Q   Median       3Q      Max
-2.21541 -0.23867  0.03459  0.28459  0.99393

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.02953    0.11209  35.949  < 2e-16 ***
cocoa_percent -0.01163    0.00156  -7.456 1.22e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4406 on 2528 degrees of freedom
Multiple R-squared:  0.02152,   Adjusted R-squared:  0.02113
F-statistic: 55.59 on 1 and 2528 DF,  p-value: 1.218e-13
```

Interpretation:

A simple linear regression was performed to examine the relationship between cocoa percentage and chocolate rating. The results indicate a statistically significant negative relationship ($\beta = -0.01163$, $p < 0.001$). On average, for each 1% increase in cocoa content, the rating decreases by approximately 0.012 points. However, the model explains only 2.15% of the variance in ratings ($R^2 = 0.0215$), suggesting that while cocoa percentage has a measurable effect, other factors likely play a much larger role in determining chocolate ratings.

7.2 Hypothesis Test: 1

Question-2: Is the mean rating significantly different between chocolates with $\geq 70\%$ cocoa and those with <70% cocoa?

- Null Hypothesis H_0: There is no difference in mean ratings between high- and low-cocoa chocolates.
- Alternative Hypothesis H_1: Mean ratings differ between the two groups.

Test and Decision ($\alpha = 0.05$):

```
    Welch Two Sample t-test
```

```
data:  rating by cocoa_group
t = 0.035992, df = 534.89, p-value = 0.9713
alternative hypothesis: true difference in means between group High Cocoa ( 70%) and group Lo
95 percent confidence interval:
 -0.04729402  0.04905940
sample estimates:
mean in group High Cocoa ( 70%)  mean in group Low Cocoa (<70%)
                       3.196479                        3.195596
```

Result: Since p-value = 0.9713, p >0.05, we accepted null hypothesis.

Interpretation:

p-value is very large (0.9713): This is much higher than any common significance level (0.05), that's mean Null Hypothesis (H_0) is accepted, there is no statistically significant difference between the mean ratings of high-cocoa and low-cocoa chocolates. Chocolates with 70% cocoa do not have a significantly different average rating compared to chocolates with <70% cocoa.

7.3 Hypothesis Test: 2

Question-3: Do chocolates from South America have higher ratings than chocolates from Africa?

H_0: $\mu_1 = \mu_2$ (no difference between the two regions)

H_1: μ_1 != μ_2 (difference exists)

Test and Decision ($\alpha = 0.05$):

```
    Welch Two Sample t-test

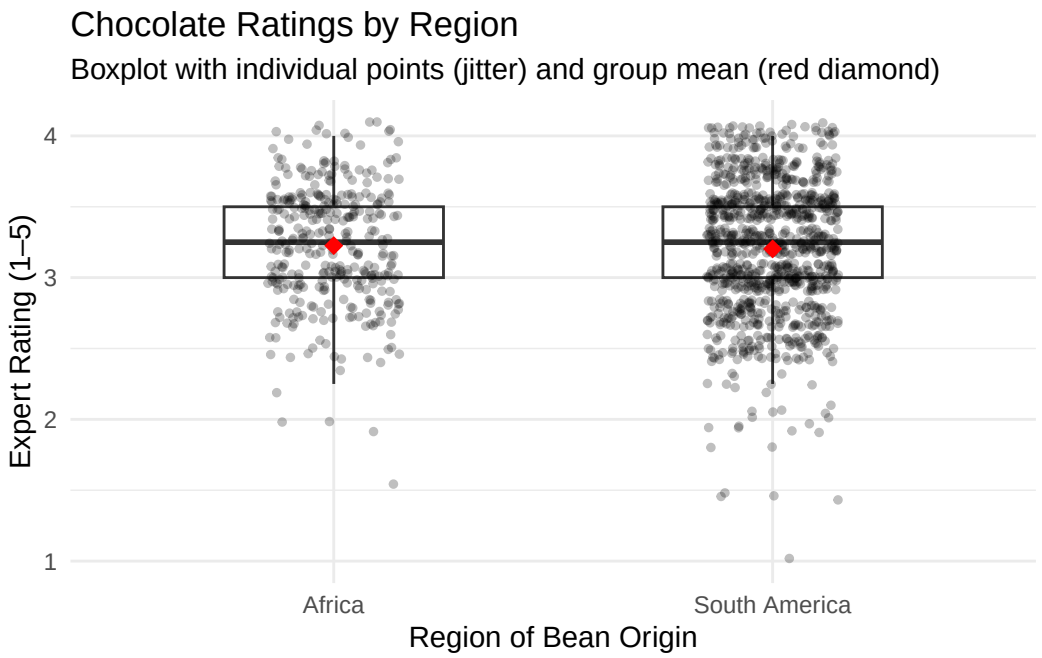
data:  rating by region
t = 0.84929, df = 678.46, p-value = 0.396
alternative hypothesis: true difference in means between group Africa and group South America
95 percent confidence interval:
 -0.03020708  0.07625860
sample estimates:
       mean in group Africa mean in group South America
                   3.225806                    3.202781
```

Result: Since p-value = 0.396, p >0.05, we accepted null hypothesis.

Interpretation:

We are comparing chocolate ratings between Africa and South America found no statistically significant difference between the two regions (t = 0.849, df = 678.46, p = 0.396). Although chocolates from Africa had a slightly higher mean rating (3.2258) compared to South America (3.2028), the difference was very small and not statistically meaningful. The 95% confidence interval for the difference (–0.0302 to 0.0763) includes zero, indicating that the true mean difference may be negligible. Overall, the test provides no evidence that chocolates from Africa and South America differ in their expert ratings.

Visualizations: Boxplot (ratings by region):



8. Limitations:

Ratings are subjective and may vary across reviewers. The dataset is not a random sample of all chocolates; premium brands are likely over-represented. Potential nonlinear effects (e.g., optimal cocoa range) were not modeled but could be explored in future work.

9. Conclusion:

We have explored how cocoa percentage and bean origin influence expert chocolate ratings using the Chocolate Ratings dataset. Across all methods—including descriptive analysis, linear regression, and hypothesis testing—the results consistently show that the variables examined have limited impact on overall chocolate ratings.

First, the simple linear regression revealed a statistically significant but very weak negative relationship between cocoa percentage and rating (β = –0.01163, p < 0.001). While higher cocoa content is slightly associated with lower ratings, the effect size is small, and cocoa percentage explains only 2.15% of the variance in expert ratings. This suggests that factors other than cocoa content—such as flavor notes, texture, brand reputation, or production quality—likely play a much larger role in determining how chocolates are rated.

Second, the comparison of chocolates with high cocoa content ($\geq$ 70%) versus those with lower content (<70%) showed no significant difference in mean ratings (p = 0.9713). As indicated in the t-test results, the mean ratings for the two groups were nearly identical. This further supports the conclusion that cocoa percentage alone is not a strong determinant of expert evaluation.

Third, the regional comparison between chocolates produced from African versus South American cocoa beans also showed no statistically significant difference (p = 0.396), as presented on page 10 of the report . Although African-origin chocolates had a slightly higher mean rating (3.2258) compared to South America (3.2028), the difference was small, and the confidence interval included zero. This indicates that, within this dataset, bean origin does not meaningfully influence expert ratings.

Overall, while the dataset offers rich information on chocolate characteristics, the analyses demonstrate that neither cocoa percentage nor bean origin strongly explains variation in expert ratings. Chocolate quality is likely influenced by a combination of more complex factors not fully captured in the dataset.

Acknowledgments:

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Appendix A: R Code Used in Analysis

A1. Loading Packages and Data

```
library(tidyverse)
library(knitr)
library(ggplot2)
library(car)
library(readr)
library(dplyr)

# Load Chocolate Ratings dataset
tuesdata <- tt_load('2022-01-18')
chocolate <- tuesdata$chocolate
```

A2. Data Cleaning

```
# Convert cocoa_percent to numeric
chocolate <- chocolate %>%
  mutate(cocoa_percent = as.numeric(str_remove(cocoa_percent, "%")))
```

A3. Variable Descriptions

```
variables <- tibble(
Variable = c("company_manufacturer", "review_date", "country_of_bean_origin",
"broad_origin", "cocoa_percent", "rating", "bean_type", "company_location"),
Description = c("Name of the company that made the chocolate bar",
"Year the chocolate bar was reviewed",
"Country where the cocoa beans were grown",
"Broader region of bean origin",
"Cocoa content of the chocolate bar, expressed as a percentage",
"Expert rating score (1.0-5.0 scale)",
"Variety or hybrid of cocoa bean used",
"Country where the manufacturer is based"),
Type = c("Categorical", "Numeric (integer)", "Categorical",
"Categorical", "Numeric (continuous)", "Numeric (continuous)",
"Categorical", "Categorical"),
```

```
Example = c("A. Morin", "2019", "Peru", "South America", "70", "3.75", "Trinitario", "France
Use_in_Analysis = c("Context variable", "Time variable", "Alternative predictor",
"Predictor variable", "Main predictor (X)", "Response variable (Y)",
"Context variable", "Optional predictor")
)
kable(variables, caption = "Variable Descriptions for Chocolate Ratings Dataset")
```

A4. Basic summary about the dataset

```
summary(chocolate$rating)

chocolate <- chocolate %>%
  mutate(cocoa_percent = as.numeric(str_replace(cocoa_percent, "%", "")))
summary(chocolate$cocoa_percent)
```

A5. Distribution of Ratings

```
ggplot(chocolate, aes(x = rating)) +
  geom_histogram(binwidth = 0.5, fill = "chocolate", color = "white") +
  labs(title = "Distribution of Chocolate Ratings", x = "Rating", y = "Count")
```

A6. Relationship Between Cocoa and Rating:

```
ggplot(chocolate, aes(x = cocoa_percent, y = rating)) +
geom_point(alpha = 0.5, color = "saddlebrown") +
geom_smooth(method = "lm", color = "darkblue", se = FALSE, formula = y ~ x) +
labs(title = "Cocoa Percentage vs Chocolate Rating",
x = "Cocoa Percentage (%)", y = "Rating")
```

A7. simple linear regression:

```
model1 <- lm(rating ~ cocoa_percent, data = chocolate)
summary(model1)
```

A8. Hypothesis Test: 1:

```
chocolate <- chocolate %>%
mutate(cocoa_group = if_else(cocoa_percent >= 70, "High Cocoa ( 70%)", "Low Cocoa (<70%)"))
t.test(rating ~ cocoa_group, data = chocolate)
```

A9. Hypothesis Test: 2:

```
south_america <- c("Peru", "Ecuador", "Venezuela", "Brazil", "Colombia", "Bolivia")
africa <- c("Madagascar", "Ghana", "Sao Tome", "Tanzania", "Uganda", "Congo")

df <- chocolate %>%
  mutate(region = case_when(
    country_of_bean_origin %in% south_america ~ "South America",
    country_of_bean_origin %in% africa ~ "Africa",
    TRUE ~ NA_character_
  )) %>%
  filter(!is.na(region))
  t.test(rating ~ region, data = df)
```

A10. Visualizations: Boxplot (ratings by region):

```
ggplot(df, aes(x = region, y = rating)) +
  geom_boxplot(outlier.shape = NA, width = 0.5) +
  geom_jitter(width = 0.15, alpha = 0.25, size = 1) +
  stat_summary(fun = mean, geom = "point", shape = 18, size = 3, color = "red") +
  labs(title = "Chocolate Ratings by Region",
       subtitle = "Boxplot with individual points (jitter) and group mean (red diamond)",
       x = "Region of Bean Origin", y = "Expert Rating (1-5)") +
  theme_minimal()
```